# Claude Code + Felix Governance Analysis

What 73 active days, 18.10 billion accounted tokens, 87,137 deduplicated model turns, and the rollout of Felix reveal about a governed autonomous engineering workflow.

**18.10B**
accounted tokens

**$16.64K**
ccusage API-equivalent cost

**Revised core finding:** the worker/subagent architecture is comparatively context-efficient, while the long-lived coordinator remains the main context-pressure point. During the commissioned Felix phase, autonomy rises sharply: 42.9 model turns and 56.9 tool calls per human prompt, with 79.4% of model turns delegated. Output per prompt rises about 60% versus pre-Felix while cost per prompt rises about 24%.

| 73 | 2,679 | 87,137 | 71.5% |
|---|---|---|---|
| active calendar days | human prompts (overall parser) | unique API/model turns | turns executed by subagents overall |

Analysis period: June 22 - September 21, 2026. Felix phase analysis is observational, not a causal experiment; model mix, projects, and task mix also changed over time.

# 1. Executive summary



**248.0M**
accounted tokens / active day

**$228**
API-equivalent cost / active day

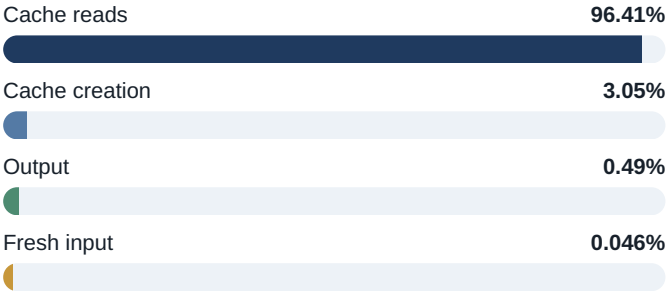
**32.5**
model turns / human prompt overall

**40.7**
tool calls / human prompt overall

> **This is not primarily a chat workflow. It is a human intent layer driving a governed, tool-using, delegated execution system.**

### What the 18.10B-token headline means

96.4% of accounted token volume is prompt-cache reads. Another 3.05% is cache creation. Fresh uncached input is only 0.046%, while generated output is 0.495%.




Cache-read to cache-creation ratio: 31.6:1.

### Overall workflow signature

**2,679 human prompts** generated about **87,137 unique model turns**. On average, one instruction launches roughly 32-33 model invocations, ~41 tool calls, and ~33K generated output tokens.

**6.72M**
context/input tokens processed per prompt

**$6.21**
API-equivalent cost per prompt overall

### Aligned 73-day accounting

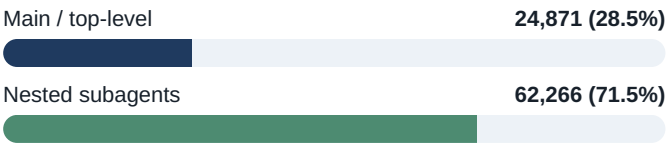
| Metric | Total | Per active day |
|---|---|---|
| Cache-read tokens | 17,451,561,051 | 239.1M |
| Cache-creation tokens | 552,851,761 | 7.57M |
| Fresh input tokens | 8,317,569 | 113.9K |
| Generated output tokens | 89,582,651 | 1.23M |
| **Total accounted tokens** | **18,102,313,032** | **248.0M** |
| ccusage API-equivalent cost | $16,641.37 | $227.96 |

# 2. Main coordinator vs. subagents

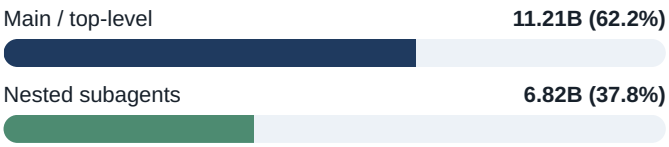


**Reconciliation quality is excellent:** globally deduplicated transcript context differs from ccusage by only +0.065%, and transcript output by +0.048%. The main/subagent split is therefore strong enough for operational conclusions.

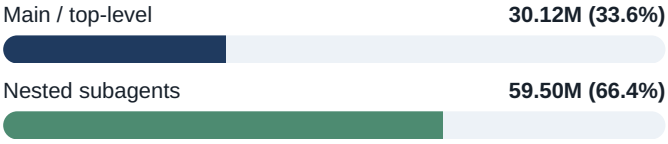
### Deduplicated model turns



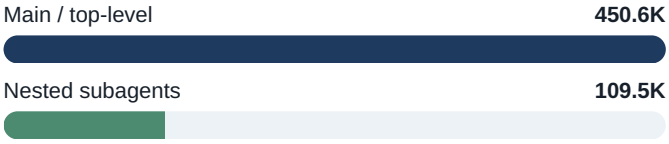

### Context processed




### Generated output




### Average context per model turn




A main-agent invocation carries about 4.12x as much context as a subagent invocation.

| Measure | Main coordinator | Subagents | Interpretation |
|---|---|---|---|
| Unique model turns | 24,871 | 62,266 | Most model calls happen in delegated work. |
| Avg. context / turn | 450,644 | 109,472 | Main turns carry 4.12x more context. |
| Context / output token | 372.1x | 114.6x | Delegated work uses far less context per generated output token. |
| Output share | 33.6% | 66.4% | Workers produce two-thirds of generated output. |

> **The worker fleet is not the obvious waste source. The architecture looks like a very large coordinator orchestrating comparatively lean workers.**

# 3. The long-context bottleneck



The largest top-level sessions repeatedly operated in the 500K-token average-context range, with p90 context commonly above 800K and peaks near 1M. High p90 values mean large context was sustained across many turns, not merely touched once.




| Session | Turns | Avg ctx | P90 ctx | Peak | Ctx/output | Cost |
|---|---|---|---|---|---|---|
| 59d07e04-b | 1,377 | 510K | 897K | 998K | 531x | $520 |
| c79569a0-3 | 1,048 | 537K | 852K | 966K | 563x | $429 |
| 7e78c467-d | 999 | 554K | 893K | 998K | 384x | $530 |
| e4c17caf-a | 940 | 524K | 872K | 998K | 798x | $324 |
| 0bcbefc4-c | 675 | 575K | 924K | 997K | 746x | $242 |

**Interpretation:** cache rebuilding is not obviously the dominant failure mode. Large cache writes are sparse relative to hundreds or thousands of turns. The costly pattern is successfully caching a huge session and repeatedly rereading it.

**Important revision:** commissioned Felix has lowered mean main-context modestly (462K pre-Felix to 436K), but p90 context has risen (767K to 827K). Felix has not yet solved long coordinator lifetimes.

## Raw opportunity if coordinator context falls

| Main-agent average context | Context avoided | Reduction vs. 18.0B context workload |
|---|---|---|
| 350K / turn | ~2.50B | ~13.9% |
| 300K / turn | ~3.75B | ~20.8% |
| 250K / turn | ~4.99B | ~27.7% |

# 4. Felix rollout: three observed phases



The local Felix evidence supports three operational phases rather than one clean switch. Felix ledger activity exists by August 11; formal commissioning records for core projects appear around September 4, followed by progressive commissioning of additional projects.

**Jun 22 - Aug 10** **Pre-Felix baseline.** Claude Code usage before observed Felix ledger activity.

**Aug 11 - Sep 3** **Early Felix.** Governance/memory/ledger behavior exists, but the later commissioning regime is still evolving.

**Sep 4 - Sep 21** **Commissioned Felix.** Formal commissioning regime is present for core projects and rolls out progressively to others.

## Workflow comparison

| Metric | Pre-Felix | Early Felix | Commissioned | Pre -> commissioned |
|---|---|---|---|---|
| Active days | 36 | 21 | 16 | - |
| Human prompts | 872 | 913 | 724 | - |
| Model turns / prompt | 32.0 | 31.1 | **42.9** | **+34.1%** |
| Tool calls / prompt | 36.2 | 40.0 | **56.9** | **+57.2%** |
| Bash calls / prompt | 22.5 | 32.0 | **43.0** | **+91.1%** |
| Subagent turn share | 69.7% | 64.6% | **79.4%** | **+9.7 pts** |
| Output / prompt | 28.3K | 35.6K | **45.2K** | **+59.7%** |
| Context / output | 244.5x | 192.0x | **177.0x** | **-27.6%** |
| Cost / prompt | $5.94 | $6.70 | **$7.39** | **+24.4%** |
| Main avg. context | 462K | 450K | **436K** | **-5.6%** |
| Main p90 context | 767K | 793K | **827K** | **+7.8%** |

**The strongest positive signal:** commissioned Felix coincides with much more work per human instruction. Output per prompt is ~60% higher, turns/prompt ~34% higher, and delegation rises to 79.4% of all model turns.

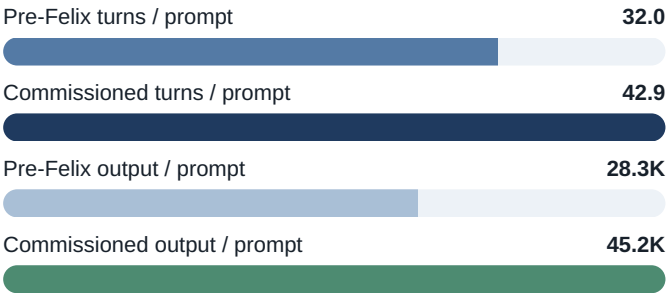
**The tradeoff:** cost per prompt rises ~24%, Bash activity nearly doubles, and p90 coordinator context is higher. Governance appears to increase execution intensity; it does not make the coordinator lightweight by itself.

# 5. What the Felix comparison suggests



> **The commissioned phase looks less like "Claude with extra rules" and more like a shift toward a governed autonomous execution system.**

### More autonomy per human instruction




### Higher spend, better token-output efficiency

Cost per prompt rises from **$5.94** to **$7.39** (+24.4%), but context/output falls from **244.5x** to **177.0x** (-27.6%).

Using generated tokens as a narrow throughput proxy, output per API-equivalent dollar improves from roughly **4.8K** to **6.1K tokens/$**, about a **28%** increase.

Generated output is not the same as engineering value; this ratio is a compute-efficiency indicator, not a quality score.

## What changed structurally

| Observed shift | Interpretation |
|---|---|
| Subagent share: 69.7% -> 79.4% | More execution is pushed into isolated worker contexts rather than retained in the coordinator. |
| Main average context: 462K -> 436K | Mean coordinator burden improves modestly, consistent with some delegation benefit. |
| Main p90 context: 767K -> 827K | Long-lived sessions still spend substantial time near the top of the context range; lifecycle management remains unresolved. |
| Tools/prompt: 36.2 -> 56.9 | The system performs far more operational work between human interventions. |
| Bash/prompt: 22.5 -> 43.0 | Execution/verification intensity is much higher; this raises both capability and operational-risk surface. |

**Causality warning:** these phases differ in more than Felix. Model mix changes substantially (for example, pre-Felix includes much more Sonnet 5, while commissioned usage is dominated by Opus 5 and Fable 5.1), and project/task mix likely changed too. The data supports "Felix-governed usage coincides with" these shifts, not "Felix alone caused" them.

**Bottom line:** the current evidence is favorable to the governance architecture. It appears compatible with more autonomy, more delegation, and better context-to-output efficiency. The clearest remaining weakness is coordinator lifespan, not worker count.

# 6. Recommended operating changes



1

### Keep Felix governance and delegation as the default for consequential work

The commissioned period is more autonomous and more delegated while producing substantially more output per human prompt. The data does not support removing the governance/worker architecture merely to reduce raw activity.

2

### Make coordinator lifecycle management a first-class Felix concern

Mean main context is slightly lower post-commissioning, but p90 is higher. Add explicit milestone-aware handoff/restart logic: persist goal, decisions, changed files, verification evidence, unresolved issues, and next actions, then start a fresh coordinator when the current one is both large and at a natural checkpoint.

3

### Track "work per human intervention" alongside cost

Cost/prompt rises, but turns/prompt, tools/prompt, and output/prompt rise faster. Report both spend and autonomy metrics so optimization does not accidentally reward a cheaper system that requires much more human steering.

4

### Separate governance overhead from productive execution

Bash calls nearly double per prompt. Instrument which calls are governance checks, verification/gates, repository inspection, tests, and task execution. This will reveal whether extra operational intensity is mostly valuable process or duplicated churn.

5

### Run task-matched experiments

The three-phase comparison is observational. For stronger evidence, compare similar tasks with and without selected Felix mechanisms, or compare matched project/task classes over time. Preserve model choice where possible so model-mix effects do not dominate.

> **Priority:** keep the governed worker fleet; engineer better coordinator handoffs and restarts; then inspect governance/tool-call overhead. The systemic optimization target is not "fewer agents."

# 7. Methodology, caveats, and final assessment

## Data sources

| Dataset | Role |
|---|---|
| ccusage aligned daily aggregate | Authoritative token categories and API-equivalent cost for June 22 - September 21. |
| Globally deduplicated Claude Code JSONL audit | 87,137 unique API message IDs; main-vs-subagent attribution based on transcript path structure. |
| Per-session context audit | Average/p90/peak context, context/output ratio, cache-rebuild candidates, and largest sessions. |
| Local Felix memory/ledger/ commissioning evidence | Phase boundaries: observed Felix ledger activity by Aug 11; formal commissioning regime around Sep 4, with progressive project rollout thereafter. |

**+0.065%**
transcript-derived context vs. ccusage

**+0.048%**
transcript-derived output vs. ccusage

## Important caveats

- **Phase comparison is observational.** Felix, model mix, project mix, user behavior, and task mix changed together.
- **Commissioning was progressive.** September 4 is a useful regime boundary for core projects, not proof that every session from that date was identically governed.
- **Felix ledger coverage is incomplete for direct session labeling.** Only four known ledger session IDs matched the analysis window, so the phase comparison is time-based rather than a perfect governed/ungoverned classifier.
- **API-equivalent cost is not necessarily the actual bill.** It is ccusage's estimate for observed token/model usage.
- **Context/output is not a quality metric.** It measures token-processing intensity relative to generated output and is best used for comparison/outlier detection.
- **Generated output is only a throughput proxy.** More output is not automatically more useful engineering work.

## Final assessment

This workflow is best described as a **governed autonomous engineering system**: the human increasingly operates at the intent/product-decision layer; a large coordinator integrates state and delegates; thousands of smaller workers perform most model turns and output; and Felix provides an evolving governance layer around execution. The commissioned phase is more autonomous and more token-output efficient, but also more operationally intense and more expensive per instruction. The strongest technical opportunity is to preserve Felix's governance and delegation benefits while giving the coordinator a much better lifecycle - durable handoffs, milestone-aware restarts, and instrumentation that distinguishes valuable verification from tool churn.

**+60%**
output / prompt, pre -> commissioned

**-27.6%**
context/output, pre -> commissioned

**+24.4%**
cost / prompt, pre -> commissioned